Circle path spaces are the integer universal-cover family

The Univalent Foundations Program

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Editorial note. Only the selected result is counted. Its local constructions and author proof are retained without generated proof completion.

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Allowed standard prerequisites (editorial).

- Univalent homotopy theory foundations. Univalence, function extensionality, dependent path transport, higher inductive pushouts/suspensions, truncations and basic pointed homotopy notions. The selected local code, fiber or gluing constructions are retained as author proof. Original numbered references identify these foundations.

Source reference. Original remark 6.10.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1187–1187. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original remark 6.10.11: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1934–1934. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original corollary 6.10.13: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1818–1818. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original lemma 6.10.12: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1885–1885. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original lemma 3.11.8: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1760–1760. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original theorem 4.7.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1891–1891. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original lemma 2.3.10: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1894–1894. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original section 2.12: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1625–1625. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original section 2.13: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1626–1626. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original section 8.1.5: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1733–1733. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original section 2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 162–162. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

0.1 $\pi_1(S^1)$

In this section, our goal is to show that $\pi_1(\mathbb{S}^1) = \mathbb{Z}$. In fact, we will show that the loop space $\Omega(\mathbb{S}^1)$ is equivalent to $\mathbb{Z}$. This is a stronger statement, because $\pi_1(\mathbb{S}^1) = \left\|\Omega(\mathbb{S}^1)\right\|_0$ by definition; so if $\Omega(\mathbb{S}^1) = \mathbb{Z}$, then $\left\|\Omega(\mathbb{S}^1)\right\|_0 = \|\mathbb{Z}\|_0$ by congruence, and $\mathbb{Z}$ is a set by definition (being a set-quotient; see original remarks 6.10.7 and 6.10.11), so $\|\mathbb{Z}\|_0 = \mathbb{Z}$. Moreover, knowing that $\Omega(\mathbb{S}^1)$ is a set will imply that $\pi_n(\mathbb{S}^1)$ is trivial for $n > 1$, so we will actually have calculated *all* the homotopy groups of $\mathbb{S}^1$.

0.1.1 Getting started

It is not too hard to define functions in both directions between $\Omega(\mathbb{S}^1)$ and $\mathbb{Z}$. By specializing original corollary 6.10.13 to $\mathsf{loop} : \mathsf{base} = \mathsf{base}$, we have a function $\mathsf{loop}^{-} : \mathbb{Z} \to (\mathsf{base} = \mathsf{base})$ defined (loosely speaking) by

$$\mathsf{loop}^n = \begin{cases} \underbrace{\mathsf{loop} \cdot \mathsf{loop} \cdot \cdots \cdot \mathsf{loop}}_{n} & \text{if } n > 0, \\ \underbrace{\mathsf{loop}^{-1} \cdot \mathsf{loop}^{-1} \cdot \cdots \cdot \mathsf{loop}^{-1}}_{-n} & \text{if } n < 0, \\ \mathsf{refl}_{\mathsf{base}} & \text{if } n = 0. \end{cases}$$

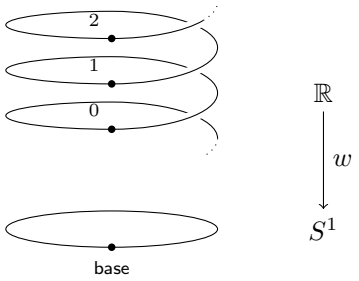


Figure 1: The winding map in classical topology

Defining a function $g : \Omega(\mathbb{S}^1) \to \mathbb{Z}$ in the other direction is a bit trickier. Note that the successor function $\mathsf{succ} : \mathbb{Z} \to \mathbb{Z}$ is an equivalence, and hence induces a path $\mathsf{ua}(\mathsf{succ}) : \mathbb{Z} = \mathbb{Z}$ in the universe $\mathcal{U}$. Thus, the recursion principle of $\mathbb{S}^1$ induces a map $c : \mathbb{S}^1 \to \mathcal{U}$ by $c(\mathsf{base}) :\equiv \mathbb{Z}$ and $\mathsf{ap}_c(\mathsf{loop}) := \mathsf{ua}(\mathsf{succ})$. Then we have $\mathsf{ap}_c : (\mathsf{base} = \mathsf{base}) \to (\mathbb{Z} = \mathbb{Z})$, and we can define $g(p) :\equiv \mathsf{transport}^{X \mapsto X}(\mathsf{ap}_c(p), 0)$.

With these definitions, we can even prove that $g(\mathsf{loop}^n) = n$ for any $n : \mathbb{Z}$, using the induction principle original lemma 6.10.12 for n. (We will prove something more general a little later on.) However, the other equality $\mathsf{loop}^{g(p)} = p$ is significantly harder. The obvious thing to try is path induction, but path induction does not apply to loops such as $p : (\mathsf{base} = \mathsf{base})$ that have *both* endpoints fixed! A new idea is required, one which can be explained both in terms of classical homotopy theory and in terms of type theory. We begin with the former.

0.1.2 The classical proof

In classical homotopy theory, there is a standard proof of $\pi_1(\mathbb{S}^1) = \mathbb{Z}$ using universal covering spaces. Our proof can be regarded as a type-theoretic version of this proof, with covering spaces appearing here as fibrations whose fibers are sets. Recall that *fibrations* over a space B in homotopy theory correspond to type families $B \to \mathcal{U}$ in type theory. In particular, for a point $x_0 : B$, the type family $(x \mapsto (x_0 = x))$ corresponds to the *path fibration* $P_{x_0}B \to B$, in which the points of $P_{x_0}B$ are paths in B starting at x_0, and the map to B selects the other endpoint of such a path. This total space $P_{x_0}B$ is contractible, since we can "retract" any path to its initial endpoint x_0 — we have seen the type-theoretic version of this as original lemma 3.11.8. Moreover, the fiber over x_0 is the loop space $\Omega(B, x_0)$ — in type theory this is obvious by definition of the loop space.

Now in classical homotopy theory, where $\mathbb{S}^1$ is regarded as a topological space, we may proceed as follows. Consider the "winding" map $w : \mathbb{R} \to \mathbb{S}^1$, which looks like a helix projecting down onto the circle (see Figure 1). This map w sends each point on the helix to the point on the circle that it is "sitting above". It is a fibration, and the fiber over each point is isomorphic to the integers. If we lift the path that goes counterclockwise around the loop on the bottom, we go up one level in the helix, incrementing the integer in the fiber. Similarly, going clockwise around the loop on the bottom corresponds to going

down one level in the helix, decrementing this count. This fibration is called the *universal cover* of the circle.

Now a basic fact in classical homotopy theory is that a map $E_1 \to E_2$ of fibrations over B which is a homotopy equivalence between E_1 and E_2 induces a homotopy equivalence on all fibers. (We have already seen the type-theoretic version of this as well in original theorem 4.7.7.) Since $\mathbb{R}$ and $P_{\mathsf{base}}S^1$ are both contractible topological spaces, they are homotopy equivalent, and thus their fibers $\mathbb{Z}$ and $\Omega(\mathbb{S}^1)$ over the basepoint are also homotopy equivalent.

0.1.3 The universal cover in type theory

Let us consider how we might express the preceding proof in type theory. We have already remarked that the path fibration of $\mathbb{S}^1$ is represented by the type family $(x \mapsto (\mathsf{base} = x))$. We have also already seen a good candidate for the universal cover of $\mathbb{S}^1$: it's none other than the type family $c : \mathbb{S}^1 \to \mathcal{U}$ which we defined in §0.1.1! By definition, the fiber of this family over base is $\mathbb{Z}$, while the effect of transporting around loop is to add one — thus it behaves just as we would expect from Figure 1.

However, since we don't know yet that this family behaves like a universal cover is supposed to (for instance, that its total space is simply connected), we use a different name for it. For reference, therefore, we repeat the definition.

Definition 0.1.1 (Universal Cover of $\mathbb{S}^1$)**.** Define $\mathsf{code} : \mathbb{S}^1 \to \mathcal{U}$ by circle-recursion, with

$$\begin{aligned} \mathsf{code}(\mathsf{base}) &:\equiv \mathbb{Z} \\ \mathsf{ap}_{\mathsf{code}}(\mathsf{loop}) &:= \mathsf{ua}(\mathsf{succ}). \end{aligned}$$

We emphasize briefly the definition of this family, since it is so different from how one usually defines covering spaces in classical homotopy theory. To define a function by circle recursion, we need to find a point and a loop in the codomain. In this case, the codomain is $\mathcal{U}$, and the point we choose is $\mathbb{Z}$, corresponding to our expectation that the fiber of the universal cover should be the integers. The loop we choose is the successor/predecessor isomorphism on $\mathbb{Z}$, which corresponds to the fact that going around the loop in the base goes up one level on the helix. Univalence is necessary for this part of the proof, because we need to convert a *non-trivial* equivalence on $\mathbb{Z}$ into an identity.

We call this the fibration of "codes", because its elements are combinatorial data that act as codes for paths on the circle: the integer n codes for the path which loops around the circle n times.

From this definition, it is simple to calculate that transporting with code takes loop to the successor function, and loop^{-1} to the predecessor function:

Lemma 0.1.2. $\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}, x) = x + 1$ *and* $\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^{-1}, x) = x - 1$.

Proof. For the first equation, we calculate as follows:

$$\begin{aligned} \mathsf{transport}^{\mathsf{code}}(\mathsf{loop}, x) &= \mathsf{transport}^{A \mapsto A}((\mathsf{code}(\mathsf{loop})), x) \\ &\qquad \text{(by original lemma 2.3.10)} \\ &= \mathsf{transport}^{A \mapsto A}(\mathsf{ua}(\mathsf{succ}), x) \quad \text{(by computation for } \mathsf{rec}_{\mathbb{S}^1}) \\ &= x + 1. \qquad \text{(by computation for } \mathsf{ua}) \end{aligned}$$

The second equation follows from the first, because $\mathsf{transport}^B(p, -)$ and $\mathsf{transport}^B(p^{-1}, -)$ are always inverses, so $\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^{-1}, -)$ must be the inverse of succ. □

We can now see what was wrong with our first approach: we defined f and g only on the fibers $\Omega(\mathbb{S}^1)$ and $\mathbb{Z}$, when we should have defined a whole morphism *of fibrations* over $\mathbb{S}^1$. In type theory, this means we should have defined functions having types

$$\prod_{x:\mathbb{S}^1} ((\mathsf{base} = x) \to \mathsf{code}(x)) \qquad \text{and/or} \tag{0.1.3}$$

$$\prod_{x:\mathbb{S}^1} (\mathsf{code}(x) \to (\mathsf{base} = x)) \tag{0.1.4}$$

instead of only the special cases of these when x is base. This is also an instance of a common observation in type theory: when attempting to prove something about particular inhabitants of some inductive type, it is often easier to generalize the statement so that it refers to *all* inhabitants of that type, which we can then prove by induction. Looked at in this way, the proof of $\Omega(\mathbb{S}^1) = \mathbb{Z}$ fits into the same pattern as the characterization of the identity types of coproducts and natural numbers in original sections 2.12 and 2.13.

At this point, there are two ways to finish the proof. We can continue mimicking the classical argument by constructing (0.1.3) or (0.1.4) (it doesn't matter which), proving that a homotopy equivalence between total spaces induces an equivalence on fibers, and then that the total space of the universal cover is contractible. The first type-theoretic proof of $\Omega(\mathbb{S}^1) = \mathbb{Z}$ followed this pattern; we call it the *homotopy-theoretic* proof.

Later, however, we discovered that there is an alternative proof, which has a more type-theoretic feel and more closely follows the proofs in original sections 2.12 and 2.13. In this proof, we directly construct both (0.1.3) and (0.1.4), and prove that they are mutually inverse by calculation. We will call this the *encode-decode* proof, because we call the functions (0.1.3) and (0.1.4) *encode* and *decode* respectively. Both proofs use the same construction of the cover given above. Where the classical proof induces an equivalence on fibers from an equivalence between total spaces, the encode-decode proof constructs the inverse map (*decode*) explicitly as a map between fibers. And where the classical proof uses contractibility, the encode-decode proof uses path induction, circle induction, and integer induction. These are the same tools used to prove contractibility — indeed, path induction *is* essentially contractibility of the path fibration composed with $\mathsf{transport}$ — but they are applied in a different way.

Since this is a book about homotopy type theory, we present the encode-decode proof first. A homotopy theorist who gets lost is encouraged to skip to the homotopy-theoretic proof (original section 8.1.5).

0.1.4 The encode-decode proof

We begin with the function (0.1.3) that maps paths to codes:

Definition 0.1.5. Define $\mathsf{encode} : \prod_{(x:\mathbb{S}^1)}(\mathsf{base} = x) \to \mathsf{code}(x)$ by

$$\mathsf{encode}\, p :\equiv \mathsf{transport}^{\mathsf{code}}(p, 0)$$

(we leave the argument x implicit).

Encode is defined by lifting a path into the universal cover, which determines an equivalence, and then applying the resulting equivalence to 0. The interesting thing about this function is that it computes a concrete number from a loop on the circle, when this loop is represented using the abstract groupoidal framework of homotopy type theory. To gain an intuition for how it does this, observe that by the above lemmas, $\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}, x)$ is the successor map and $\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^{-1}, x)$ is the predecessor map. Further, $\mathsf{transport}$ is functorial (original section 2), so $\mathsf{transport}^{\mathsf{code}}(\mathsf{loop} \cdot \mathsf{loop}, -)$ is

$$(\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}, -)) \circ (\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}, -))$$

and so on. Thus, when p is a composition like

$$\mathsf{loop} \cdot \mathsf{loop}^{-1} \cdot \mathsf{loop} \cdot \cdots$$

$\mathsf{transport}^{\mathsf{code}}(p, -)$ will compute a composition of functions like

$$\mathsf{succ} \circ \mathsf{pred} \circ \mathsf{succ} \circ \cdots$$

Applying this composition of functions to 0 will compute the *winding number* of the path — how many times it goes around the circle, with orientation marked by whether it is positive or negative, after inverses have been canceled. Thus, the computational behavior of encode follows from the reduction rules for higher-inductive types and univalence, and the action of $\mathsf{transport}$ on compositions and inverses.

Note that the instance $\mathsf{encode}' :\equiv \mathsf{encode}_{\mathsf{base}}$ has type $(\mathsf{base} = \mathsf{base}) \to \mathbb{Z}$. This will be one half of our desired equivalence; indeed, it is exactly the function g defined in §0.1.1.

Similarly, the function (0.1.4) is a generalization of the function loop^- from §0.1.1.

Definition 0.1.6. Define $\mathsf{decode} : \prod_{(x:\mathbb{S}^1)} \mathsf{code}(x) \to (\mathsf{base} = x)$ by circle induction on x. It suffices to give a function $\mathsf{code}(\mathsf{base}) \to (\mathsf{base} = \mathsf{base})$, for which we use loop^-, and to show that loop^- respects the loop.

Proof. To show that loop^- respects the loop, it suffices to give a path from loop^- to itself that lies over loop. By the definition of dependent paths, this means a path from

$$\mathsf{transport}^{(x' \mapsto \mathsf{code}(x') \to (\mathsf{base} = x'))}(\mathsf{loop}, \mathsf{loop}^-)$$

to loop^-. We define such a path as follows:

$$\begin{aligned}
&\mathsf{transport}^{(x' \mapsto \mathsf{code}(x') \to (\mathsf{base} = x'))}(\mathsf{loop}, \mathsf{loop}^-) \\
&\quad = \mathsf{transport}^{x' \mapsto (\mathsf{base} = x')}(\mathsf{loop}) \circ \mathsf{loop}^- \circ \mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^{-1}) \\
&\quad = (- \cdot \mathsf{loop}) \circ (\mathsf{loop}^-) \circ \mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^{-1}) \\
&\quad = (- \cdot \mathsf{loop}) \circ (\mathsf{loop}^-) \circ \mathsf{pred} \\
&\quad = (n \mapsto \mathsf{loop}^{n-1} \cdot \mathsf{loop}).
\end{aligned}$$

On the first line, we apply the characterization of $\mathsf{transport}$ when the outer connective of the fibration is $\to$, which reduces the $\mathsf{transport}$ to pre- and post-composition with $\mathsf{transport}$ at the domain and codomain types. On the second

line, we apply the characterization of $\mathsf{transport}$ when the type family is $x \mapsto \mathsf{base} = x$, which is post-composition of paths. On the third line, we use the action of code on loop^{-1} from Lemma 0.1.2. And on the fourth line, we simply reduce the function composition. Thus, it suffices to show that for all n, $\mathsf{loop}^{n-1} \cdot \mathsf{loop} = \mathsf{loop}^n$. This is an easy application of original lemma 6.10.12, using the groupoid laws. □

We can now show that encode and decode are quasi-inverses. What used to be the difficult direction is now easy!

Lemma 0.1.7. *For all $x : \mathbb{S}^1$ and $p : \mathsf{base} = x$, $\mathsf{decode}_x(\mathsf{encode}_x(p)) = p$.*

Proof. By path induction, it suffices to show that $\mathsf{decode}_{\mathsf{base}}(\mathsf{encode}_{\mathsf{base}}(\mathsf{refl}_{\mathsf{base}})) = \mathsf{refl}_{\mathsf{base}}$. But $\mathsf{encode}_{\mathsf{base}}(\mathsf{refl}_{\mathsf{base}}) \equiv \mathsf{transport}^{\mathsf{code}}(\mathsf{refl}_{\mathsf{base}}, 0) \equiv 0$, and $\mathsf{decode}_{\mathsf{base}}(0) \equiv \mathsf{loop}^0 \equiv \mathsf{refl}_{\mathsf{base}}$. □

The other direction is not much harder.

Lemma 0.1.8. *For all $x : \mathbb{S}^1$ and $c : \mathsf{code}(x)$, we have $\mathsf{encode}_x(\mathsf{decode}_x(c)) = c$.*

Proof. The proof is by circle induction. It suffices to show the case for base, because the case for loop is a path between paths in $\mathbb{Z}$, which is immediate because $\mathbb{Z}$ is a set.

Thus, it suffices to show, for all $n : \mathbb{Z}$, that

$$\mathsf{encode}'(\mathsf{loop}^n) = n.$$

The proof is by induction, using original lemma 6.10.12.

- In the case for 0, the result is true by definition.
- In the case for $n + 1$,

$$\begin{aligned}
\mathsf{encode}'(\mathsf{loop}^{n+1}) &= \mathsf{encode}'(\mathsf{loop}^n \cdot \mathsf{loop}) && \text{(by definition of } \mathsf{loop}^-\text{)} \\
&= \mathsf{transport}^{\mathsf{code}}((\mathsf{loop}^n \cdot \mathsf{loop}), 0) \\
& && \text{(by definition of } \mathsf{encode}\text{)} \\
&= \mathsf{transport}^{\mathsf{code}}(\mathsf{loop}, (\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^n, 0))) \\
& && \text{(by functoriality)} \\
&= (\mathsf{transport}^{\mathsf{code}}(\mathsf{loop}^n, 0)) + 1 && \text{(by Lemma 0.1.2)} \\
&= n + 1. && \text{(by the inductive hypothesis)}
\end{aligned}$$

- The case for negatives is analogous. □

Finally, we conclude the theorem.

Theorem 0.1.9. *There is a family of equivalences $\prod_{(x:\mathbb{S}^1)}((\mathsf{base} = x) \simeq \mathsf{code}(x))$.*

Proof. The maps encode and decode are quasi-inverses by Lemmas 0.1.7 and 0.1.8. □